Fine Time Sync v0.2

Performance & Stress Tests

22/Dec/2025

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Github: https://github.com/abbbe/fts/

Content of This Presentation

- Part I – Introduction
  - Why and how sync clocks and discipline timers
  - Demo 1: free running clock
  - Demo 2: disciplined clocks
- Part II - New Setup
  - SDR-based “self-validating” test setup
  - Demo 3: validate the setup
  - Demo 4: measure FTS performance
- Part III - push FTS to the limit

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  - Demo 5: increase distance, add walls
  - Demo 6: saturate the wifi link

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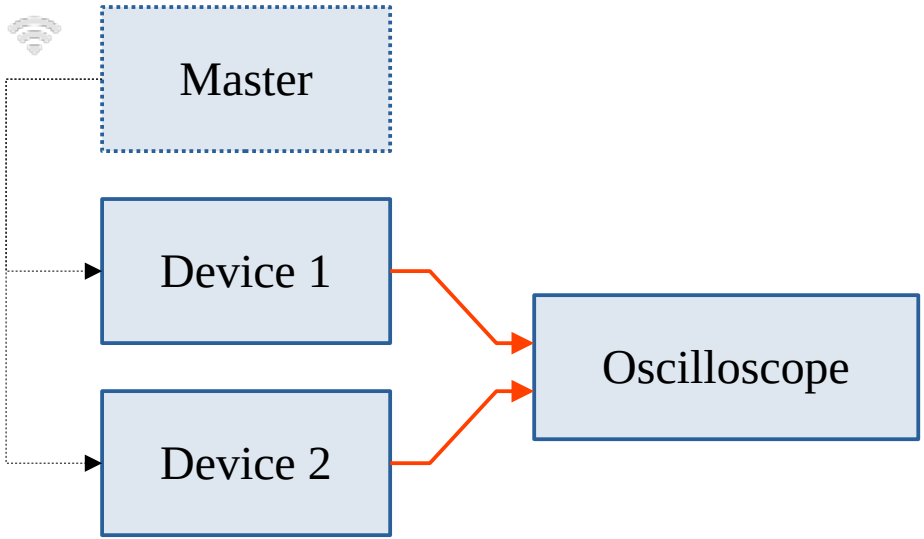
Part I

Introduction

Why and How

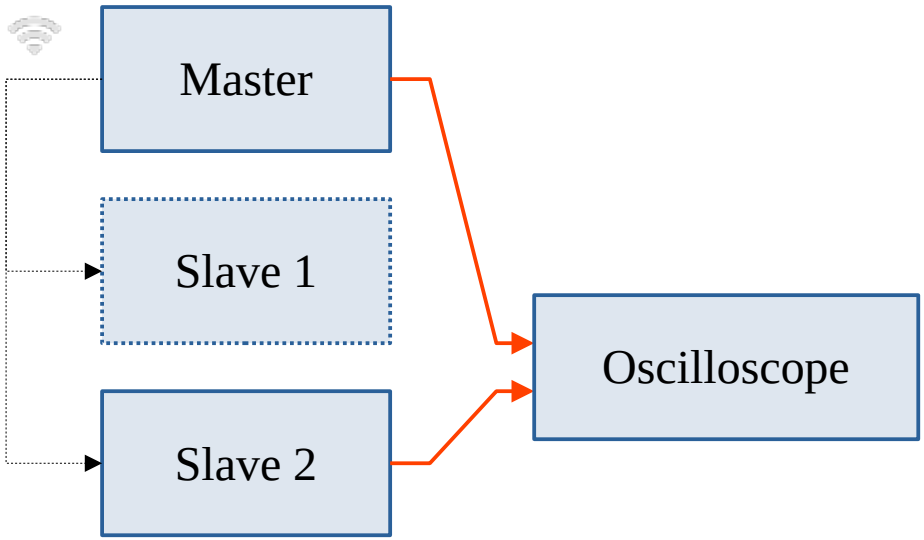
- Synced clocks are useful
  - Correlate sensor readings (convert between local and remote clocks)
  - Take synchronous actions (synchronized / disciplined timers)
- Independent clocks
  - Need initial sync
  - Diverge rapidly if left uncorrected
- Common solutions:
  - NTP: ~10-100us, wired or wireless
  - PTP: ~10-100ns, wired, special hardware
  - Disciplined timers … none?
- **FTS: <100ns (99%)**, ESP32 S3 Wifi FTM

Demo 1: Free Running Clocks



- Clocks are not synced
- Clocks sync (one)
- Clocks diverge
- Impact of capacitance

Demo 2: Disciplined Clocks



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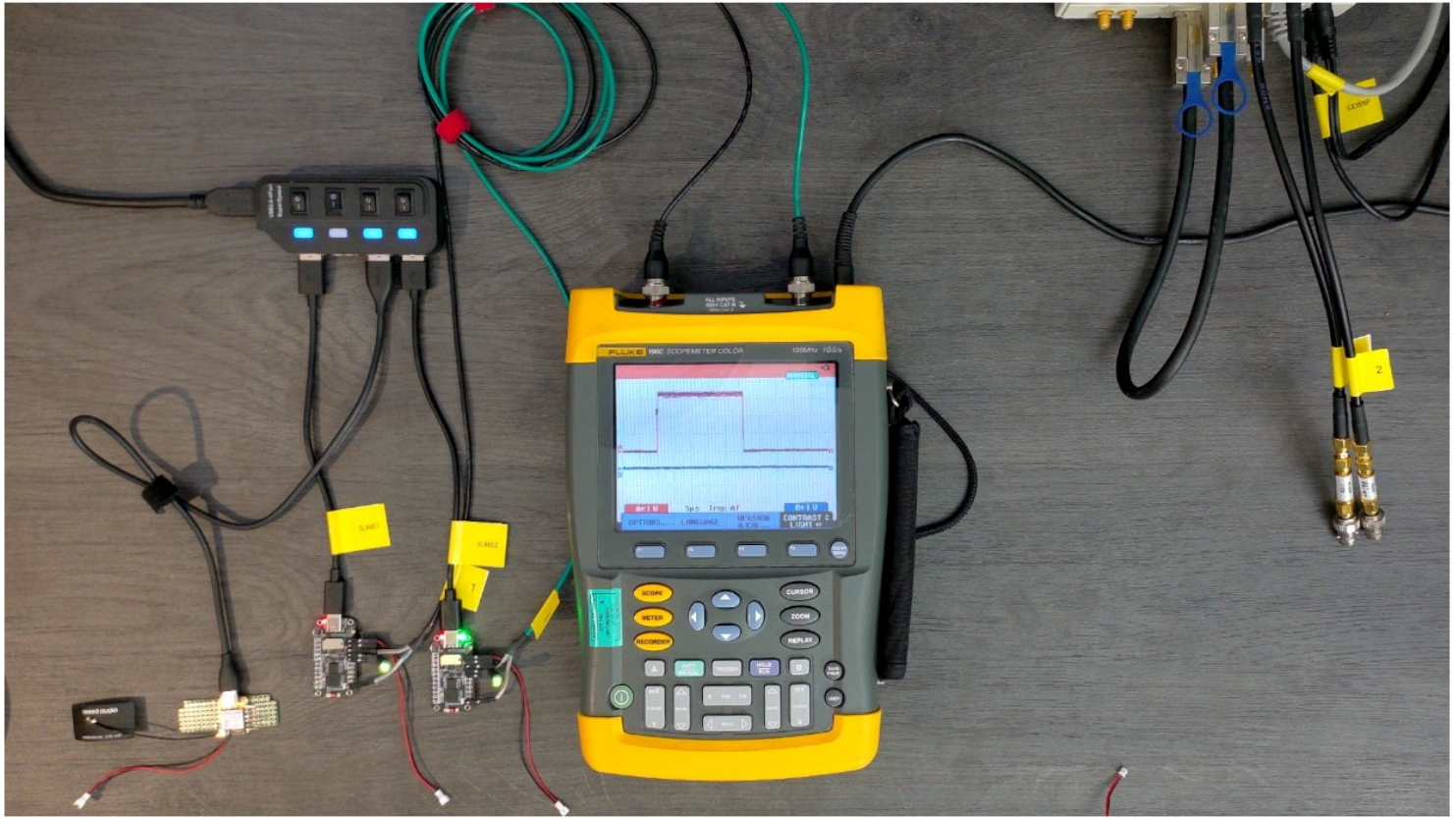
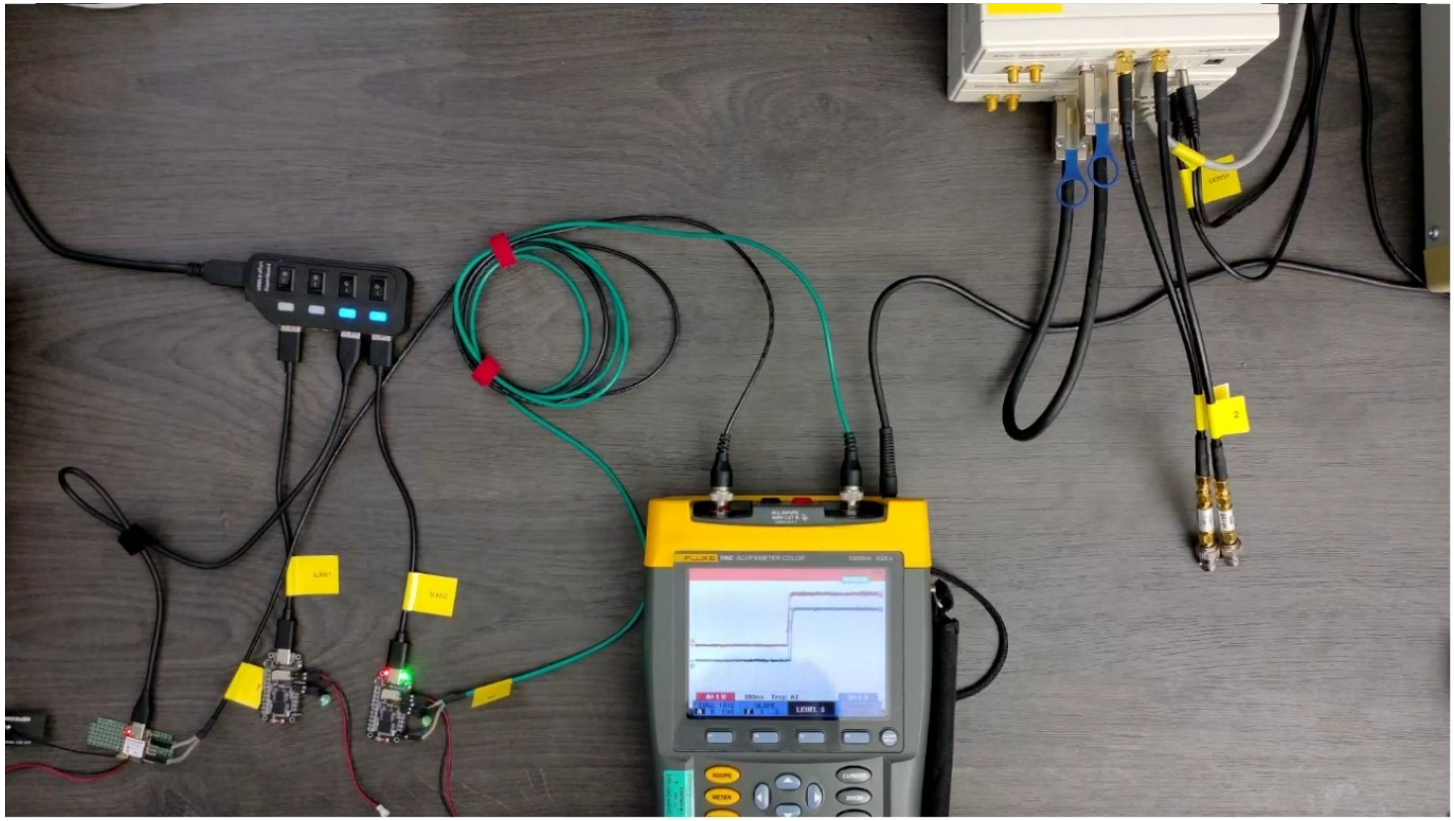


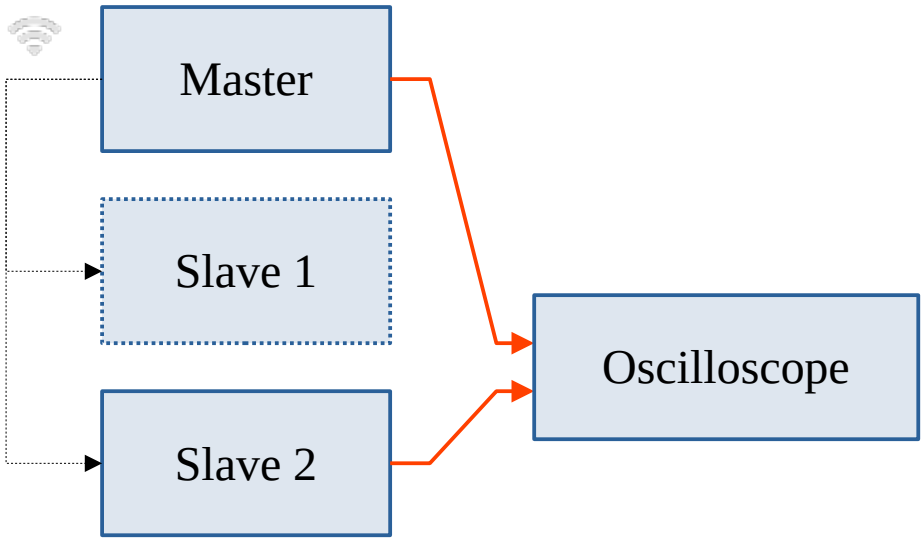
Demo2: Disciplined Clocks



Part II

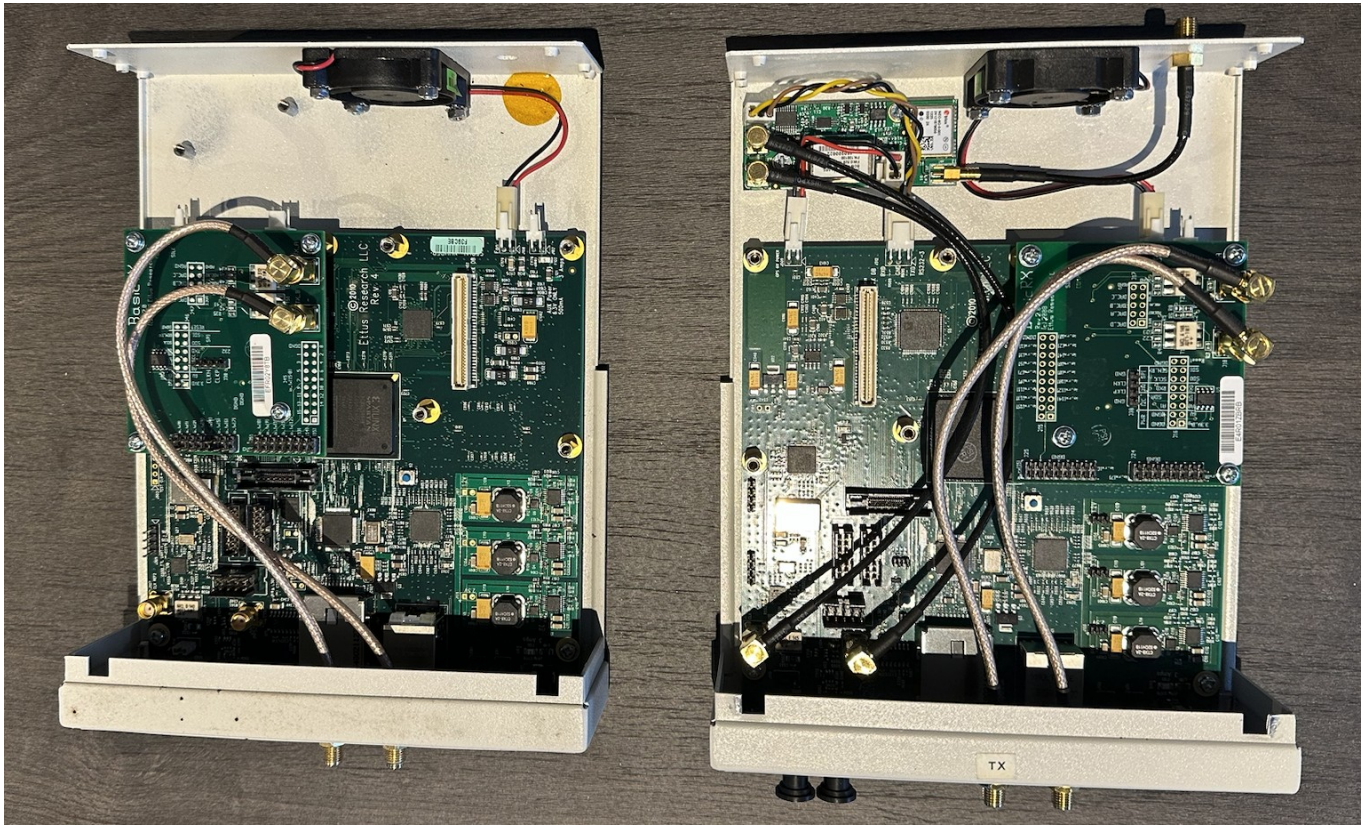
SDR-based “Self-Validating” Setup

Problems of (my) Scope-based Setup

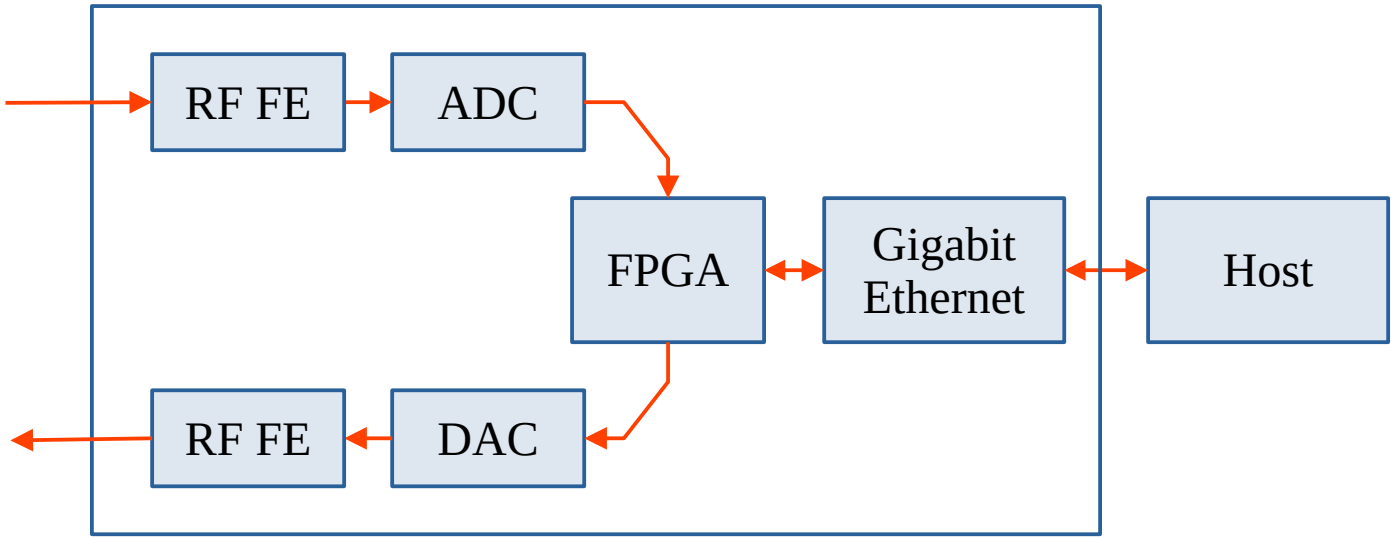


- Can miss glitches
- No long term capture
- Difficult to quantify

2x Ettus N210 + Basic TX / RX + GPSDO

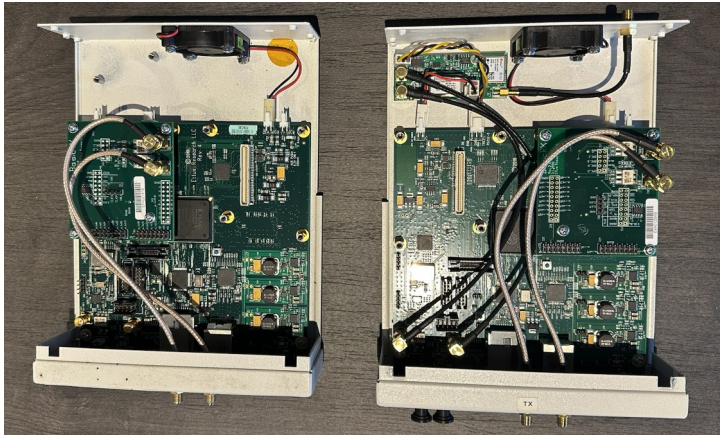


SDR (Software Defined Radio)

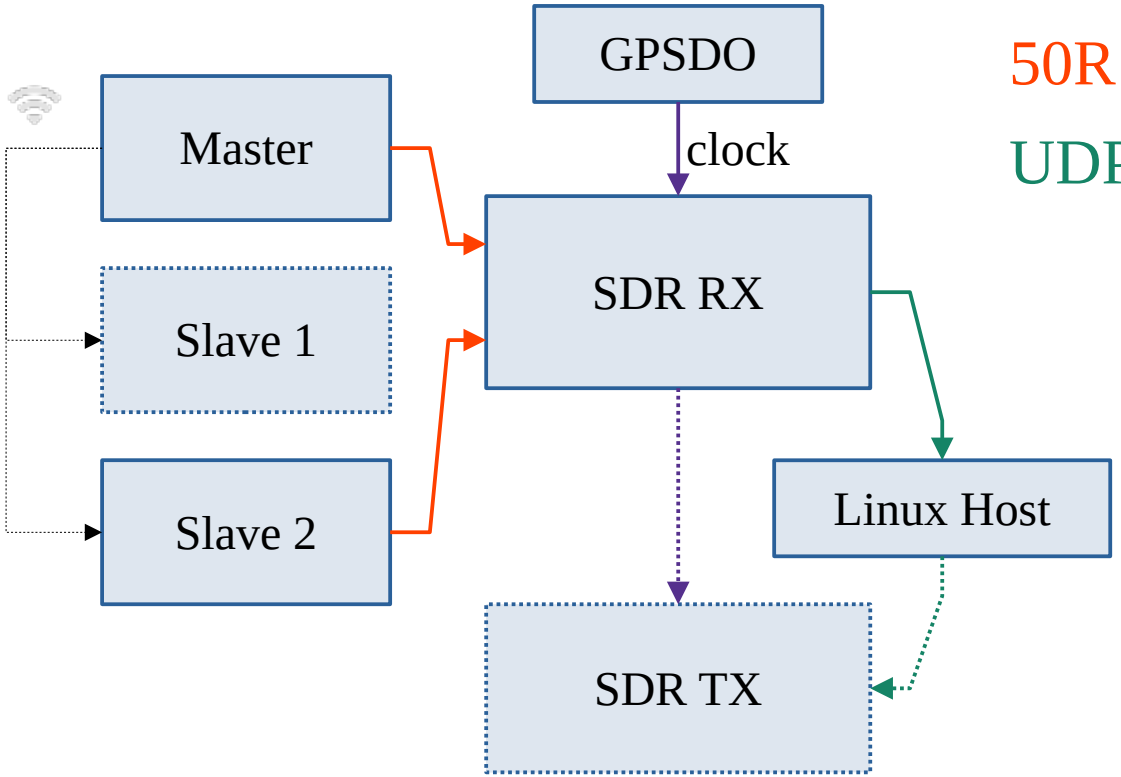


Ettus N210 SDRs

- Ettus USRP N210 - well known lab-grade SDRs
- 2 units: one with Basic RX, another with Basic TX board
- **2x Rx + 2x TX channels**, 100MSps (total, theoretical)
- 1-250MHz bandwidth
- Shared clock (MIMO cable)
- **GPS disciplined oscillator**
- UDP over Gigabit Ethernet
- **Python UHD API**



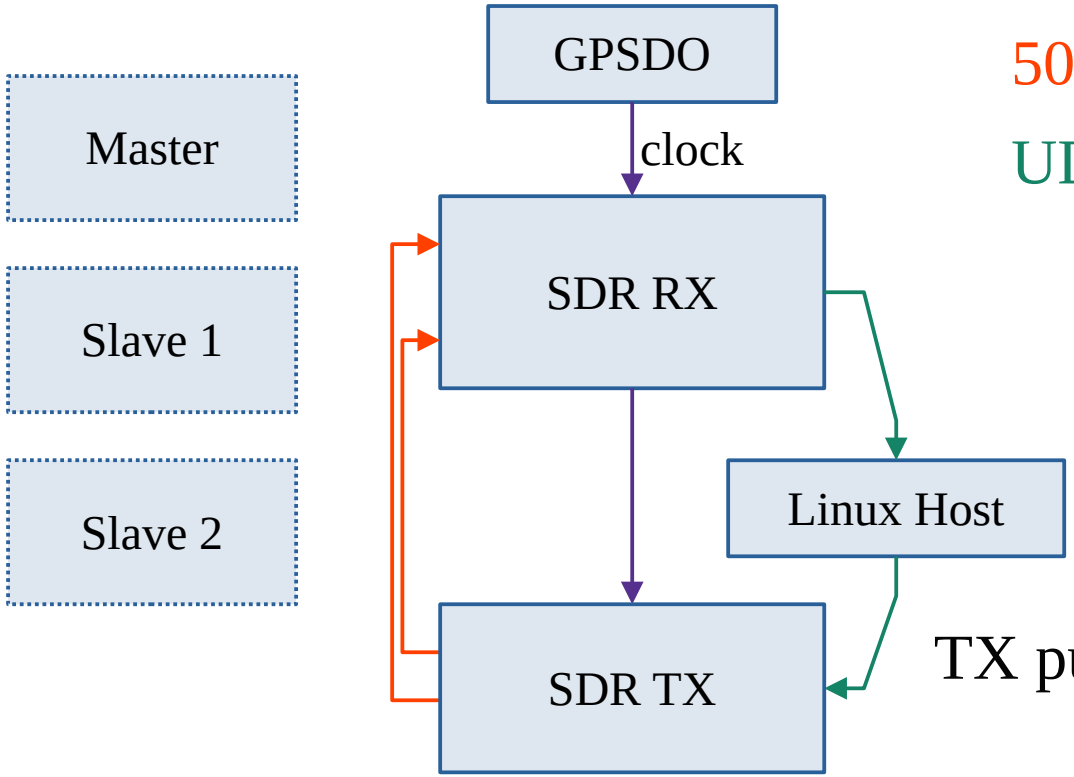
The New Setup



50R coax in

UDP over 1GE out

“Self-validation” mode

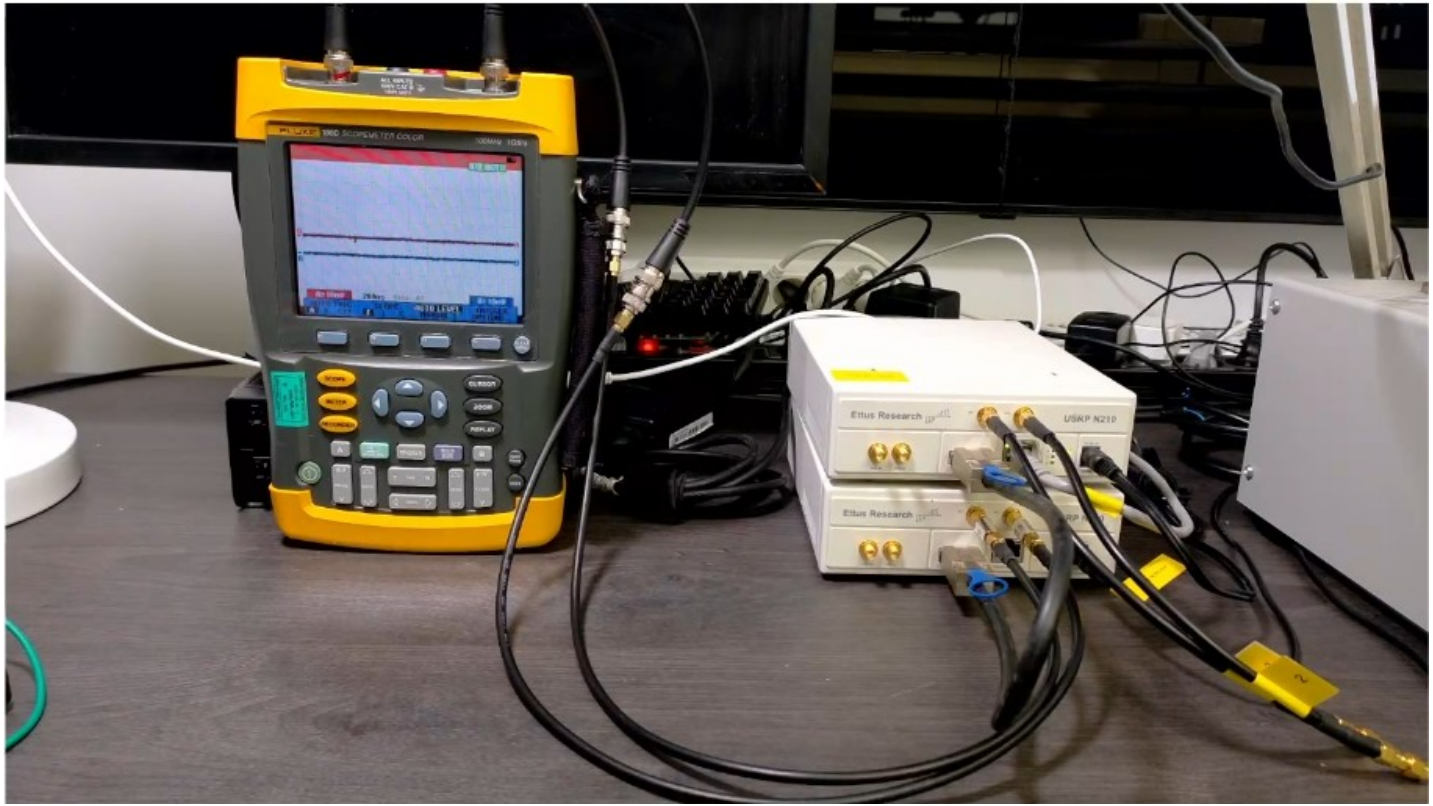


50R coax in / out

UDP over 1GE out / in

TX pulses with known jitter,
RX, measure, compare

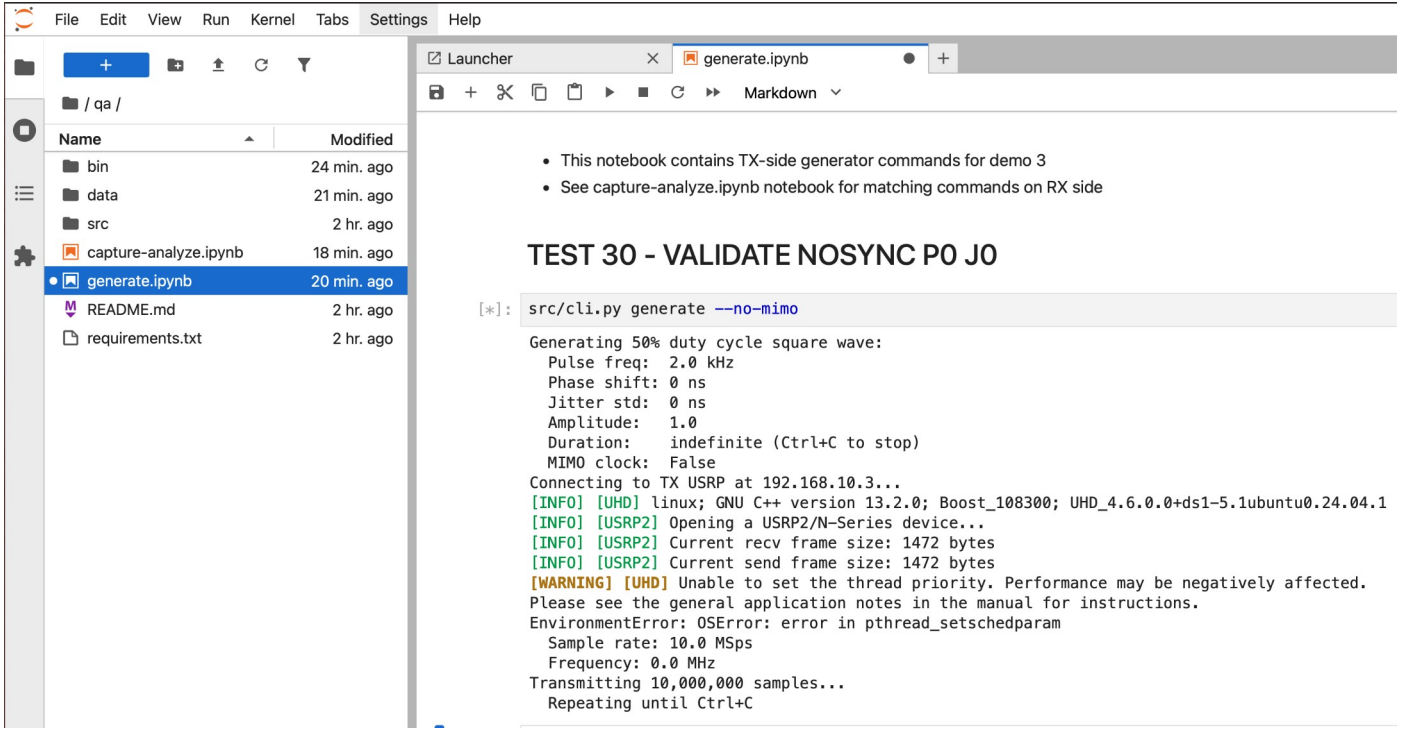
Demo 3a: Sanity Check (TX SDR → scope)



Validation (TX SDR → RX SDR)

- We will run 5 tests:
  1) 0 shift, 0 jitter (without TX/RX sync)
  2) 0 shift, 0 jitter
  3) +1us shift +/-250ns jitter
  4) -20ns shift +/-20ns jitter
  5) -20ns shift +/-20ns jitter (8h)
- Expected accuracy ~ 10ns

Demo 3b: TX SDR → RX SDR



0 shift, 0 jitter (no TX/RX sync)

FTS-QA Analysis Report

Input: data/test30-validate-nosync-p+0-j+0
Detection: threshold-crossing
Generated: 2025-12-13 23:53:11

Jitter Statistics

Count	1,219,600
Mean	+1.929 ns
Std Dev	0.375 ns
Min	+1.166 ns
Max	+3.496 ns
P50	±0.236 ns
P95	±0.774 ns
P99	±1.108 ns

Channel A Period

Mean Period	491.883 µs
Std Dev	8964.503 µs
Frequency	2033.004605 Hz
Error from Nominal	+16234.4 ppm

Channel B Period

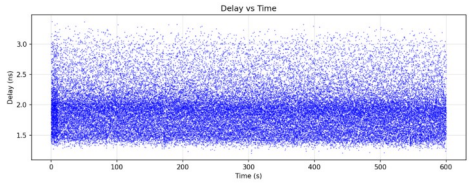
Mean Period	491.883 µs
Std Dev	8964.503 µs
Frequency	2033.004605 Hz
Error from Nominal	+16234.4 ppm

Frequency Skew

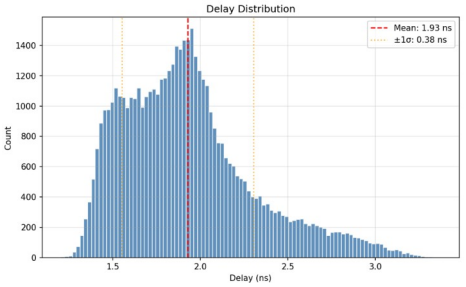
Skew (B vs A)	+0.0000 ppm
Drift Rate	+0.0 ns/s

Delay Analysis

Delay vs Time



Delay Distribution



0 shift, 0 jitter

FTS-QA Analysis Report

Input: data/test30-validate-sync-p+0-j+0
Detection: threshold-crossing
Generated: 2025-12-08 01:06:28

Jitter Statistics

Count	119,800
Mean	+1.968 ns
Std Dev	0.055 ns
Min	+1.734 ns
Max	+2.190 ns
P50	±0.036 ns
P95	±0.107 ns
P99	±0.139 ns

Channel A Period

Mean Period	500.000 µs
Std Dev	0.000 µs
Frequency	2000.000000 Hz
Error from Nominal	+0.0 ppm

Channel B Period

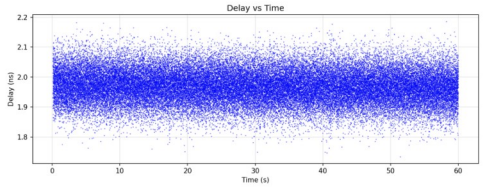
Mean Period	500.000 µs
Std Dev	0.000 µs
Frequency	2000.000000 Hz
Error from Nominal	+0.0 ppm

Frequency Skew

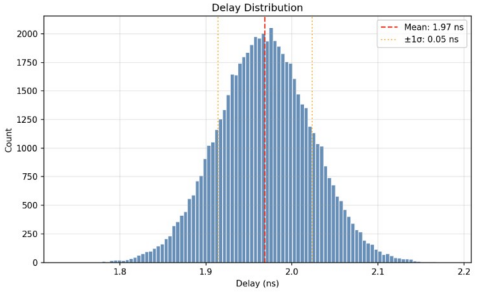
Skew (B vs A)	-0.0000 ppm
Drift Rate	-0.0 ns/s

Delay Analysis

Delay vs Time



Delay Distribution



+1us shift, +/- 250ns jitter

FTS-QA Analysis Report

Input: data/test32-validate-p+1000-j+250
Detection: threshold-crossing
Generated: 2025-12-08 13:18:14

Jitter Statistics

Count	119,800
Mean	+1001.958 ns
Std Dev	249.422 ns
Min	+139.740 ns
Max	+1799.951 ns
P50	±167.644 ns
P95	±488.466 ns
P99	±646.092 ns

Channel A Period

Mean Period	500.000 µs
Std Dev	0.000 µs
Frequency	2000.000000 Hz
Error from Nominal	+0.0 ppm

Channel B Period

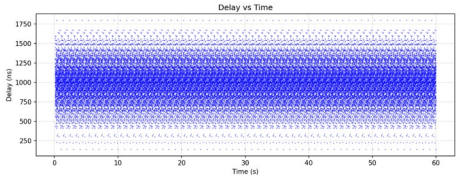
Mean Period	500.000 µs
Std Dev	0.355 µs
Frequency	2000.000015 Hz
Error from Nominal	+0.0 ppm

Frequency Skew

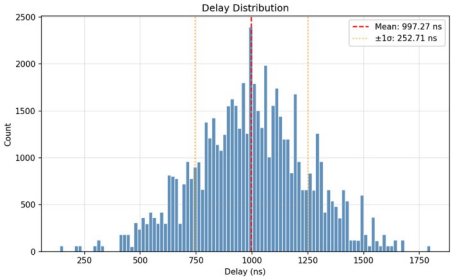
Skew (B vs A)	+0.0077 ppm
Drift Rate	+7.7 ns/s

Delay Analysis

Delay vs Time



Delay Distribution



20ns shift +/- 20ns jitter

FTS-QA Analysis Report

Input: data/test33-validate-p-20-j+20
Detection: threshold-crossing
Generated: 2025-12-08 01:09:10

Jitter Statistics

Count	119,800
Mean	-9.908 ns
Std Dev	17.807 ns
Min	-81.082 ns
Max	+61.814 ns
P50	±9.794 ns
P95	±38.729 ns
P99	±51.437 ns

Channel A Period

Mean Period	500.000 µs
Std Dev	0.000 µs
Frequency	2000.000000 Hz
Error from Nominal	+0.0 ppm

Channel B Period

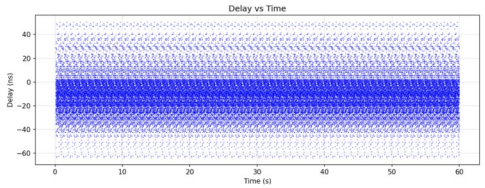
Mean Period	500.000 µs
Std Dev	0.025 µs
Frequency	1999.999999 Hz
Error from Nominal	-0.0 ppm

Frequency Skew

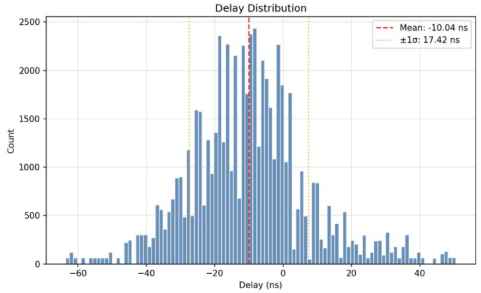
Skew (B vs A)	-0.0003 ppm
Drift Rate	-0.3 ns/s

Delay Analysis

Delay vs Time



Delay Distribution



20ns shift +/- 20ns jitter (8h capture)

FTS-QA Analysis Report

Input: data/test34-validate-p-20-j+20-long
Detection: threshold-crossing
Generated: 2025-12-08 10:06:14

Jitter Statistics

Count	57,599,156
Mean	-8.929 ns
Std Dev	16.433 ns
Min	-86.615 ns
Max	+64.516 ns
P50	±8.639 ns
P95	±35.307 ns
P99	±50.680 ns

Channel A Period

Mean Period	500.000 µs
Std Dev	0.004 µs
Frequency	1999.999998 Hz
Error from Nominal	-0.0 ppm

Channel B Period

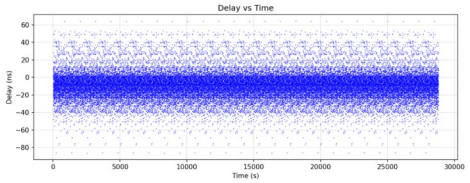
Mean Period	500.000 µs
Std Dev	0.023 µs
Frequency	1999.999998 Hz
Error from Nominal	-0.0 ppm

Frequency Skew

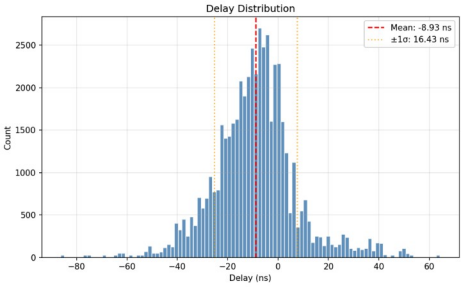
Skew (B vs A)	+0.0000 ppm
Drift Rate	+0.0 ns/s

Delay Analysis

Delay vs Time



Delay Distribution



Conclusion

- Generator → transmitter → **receiver** → **signal processing**
- Deviation from expected results ~ 10ns
- => Can trust the second half of this chain

Demo 4: Measure FTS Performance

TODO

Part III

Push FTS to the limits

Pushing FTS to the Limits

TODO